Ahmad Hossein **Yazdani**

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Github Linkedin

Summary

PhD candidate specializing in ML systems infrastructure and performance optimization for large-scale distributed training. Experienced in diagnosing and mitigating GPU, storage, and communication bottlenecks through profiling and systems-level experimentation. Focused on building scalable optimizations for production-scale AI workloads.

Work Experience

August,2020 – present **Research Assistant at Distributed System and Storage Lab, Virginia Tech**.

Advisor : Dr. Ali Butt, *Professor, Department of Computer Science*, Virginia Tech

◦ **LLMStore:** Designing caching and activation-ranking mechanisms to reduce tensor offload latency in distributed LLM training; identified 300GB+ (Llama-70B) offload traffic during LLaMA fine-tuning via trace analysis.

◦ **Silo-based open framework for Federated Learning (2025-):** A **Federated Learning research** project led by Dr. Feiyi Wang (ORNL), Dr. Ali Butt (Virginia Tech), and Dr. Sahil Tyagi (ORNL). Implemented gradient compression in the gRPC communication layer for a cross-cloud/HPC Federated Learning framework, reducing cross-site communication overhead and enabling hierarchical aggregation.

◦ **Metis** The project introduces a data access pattern for Deep Learning workloads to improve cachability. With a small cache, it improves the hit ratio by up to 4.5× compared to the LRU policy : **Shade paper at FAST23**

January,2026 – ***Research Intern, Oak Ridge National Laboratory***.

Mentors: William Godoy, Pedro Vallero

◦ Selected for ORNL research internship (2026) to explore AI-driven optimization of memory- and storage-intensive HPC workflows.

June,2024 – August,2024 ***Student Assistant at NERSC, Lawrence Berkeley National Laboratory (LBNL), internship***.

Mentors: Stephen Simms, Lisa Gerhardt, Jean Luca Bez

◦ Examined Drishti, an HPC I/O recommendation tool; Detected many false positive warnings, and derived insights to improve the accuracy of the the Drishti I/O recommendation tool.

June,2023 – August,2023 ***Student Assistant at Lawrence Berkeley National Laboratory (LBNL), internship***.

Mentors: Suren Byna, Jean Luca Bez

◦ Identified a significant I/O variability for various HPC applications like E3SM (For earth modeling) and LAMMP (For molecular simulations). Later continued my research for Large AI models and language models : **LLMStore**

June,2021 – August,2021 ***Internship at Oak Ridge National Laboratory, Analytics & AI Methods at Scale Group***.

Mentors: Feiyi Wang, Sarp Oral, Ahmad Maroof Karimi and Arnab Kamur Paul

◦ **This experience resulted in my ICDCN'25 paper, where I was able to predict the I/O pattern of the next job given the features from the past submissions of the same user with an accuracy of nearly 90% for HPC jobs.**

Computer skills

Programming Languages & frameworks	Python, R, C, C++, Advanced JAVA EE, JAVA SE, Tensorflow, Go, Rust, MPI, pthread, OpenMP, CUDA, Java bytecode, X86 assembly
ML	PyTorch, keras, scikit-learn, DeepSpeed, LLM, Computer Vision, Federated Learning
Systems	Distributed systems, File Systems, Lustre, GPFS, Slurm, MapReduce, Redis, Ceph, NVMe, Darshan, Docker

Conference & Workshop publications

[ICDCN'25] **Yazdani, Ahmad Hossein**, Arnab K. Paul, Ahmad Maroof Karimi, Feiyi Wang, and Ali Butt. User-based i/o profiling for leadership scale hpc workloads. In *Proceedings of the 26th International Conference on Distributed Computing and Networking*, ICDCN '25, page 181–190, New York, NY, USA, Jan. 2025. Association for Computing Machinery. doi:10.1145/3700838.3700865.

[**FAST'23★**] Redwan Ibne Seraj Khan, **Yazdani, Ahmad Hossein**, Yuqi Fu, Arnab K Paul, Bo Ji, Xun Jian, Yue Cheng, and Ali R Butt. Shade: Enable fundamental cacheability for distributed deep learning training. In *Proceedings of the 21th USENIX Conference on File and Storage Technologies*, page 14, Santa Clara, CA, US, Feb. 2023. USENIX Association. URL `https://www.usenix.org/conference/fast23/presentation/khan`.

★ Top-tier venue

Education

2020–August 2026	**PhD, Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US. **Advisor:** Dr Ali Butt
May 2025	**Masters of Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US. **Advisor:** Dr Ali Butt
2015–2020 :	**Bachelor of Computer Software Engineering**, *University of Tehran*, Tehran, Iran.